

Student Name:

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TRIAL HIGHER SCHOOL CERTIFICATE

MATHEMATICS

Extension 1



Question 7 (continued)

(iii) Find the rate at which the water is overflowing from the container when the vertex of the cone is 2 metres below the surface.

2

(c) (i) What is the domain of $h(x) = \sin^{-1}\sqrt{1-x^2} + \sin^{-1}x$?

1

(ii) Find $h'(x)$

2

(iii) Hence determine the interval over which $h(x)$ is constant and find this constant.

2

End of paper

Question 1 (12 Marks)

(a) Find $\int \frac{1}{\sqrt{25-x^2}} dx$

Marks

1

(b) Find the exact value of $\int_0^1 \left(\frac{1}{1+x} + e^{-x} \right) dx$

2

(c) State the domain and range of $y = \sin^{-1}\left(\frac{x}{3}\right)$

2

(d) Using the substitution $u = 2+x^2$, or otherwise, find $\int x\sqrt{2+x^2} dx$

3

(e) Find the coordinates of the point which divides the interval AB with A(1, 4) and B(5,2) externally in the ratio 1:3.

2

(f) The acute angle between the lines $y = 2x + 7$ and $y = mx - 3$ is 45° . Find the two possible values of m .

2

13

Question 2 (12 Marks) Start a new page

Marks

- (a) Find $\frac{d}{dx}(\cos^{-1} 3x)$
- (b) A particle is moving along the x -axis. Its velocity V at position x is given by $V = \sqrt{8x - x^2}$. Find the acceleration when $x = 3$.

1

2

- (c) (i) Differentiate $x - x \ln x$

2

- (ii) Hence, or otherwise, evaluate $\int_1^e \ln x \, dx$

2

- (d) A bowl of hot soup at temperature $T^\circ \text{C}$, when placed in cooler surrounding air, loses heat according to the law

$$\frac{dT}{dt} = -k(T - S)$$

where t is the time elapsed in minutes and S is the temperature of the surrounding air in degrees Celsius.

- (i) Show that $T = S + Ae^{-kt}$ satisfies this equation.
- (ii) A bowl of soup at 96° is left to stand in a room at a temperature of 18°C . After 3 minutes the soup cools to 75° . Calculate the value of k to 4 decimal places.
- (iii) Melissa wishes to enjoy her soup at a temperature of 60° . How long should she wait?

1

2

2

Question 3 (12 Marks) Start a new page.

Marks

- (a) (i) Let $g(x) = x^3 + 5x^2 + 17x - 10$. Show that $g(x) = 0$ has a root between 0 and 2. [This is the only root]
- (ii) Use one application of the "halving the interval" method to find a smaller interval containing the root.
- (iii) Which end of the smaller interval found in part (ii) is closer to the root? Briefly justify your answer.

1

1

2

- (b) Show that $\frac{d}{dx}(\sec x) = \sec x \tan x$

2

- (c) Using the substitution $x = u - 1$ find the integral $\int \frac{1}{x^2 + 2x + 4} \, dx$

2

- (d)

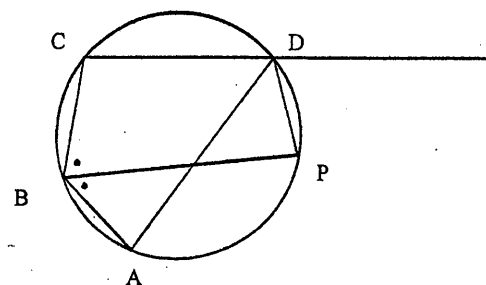


Diagram not to scale

In the diagram above ABCD is a cyclic quadrilateral. CD is produced to E. P is a point on the circle through A, B, C, D such that $\angle ABP = \angle PBC$.

- (i) Copy the diagram showing the above information.
- (ii) Explain why $\angle ABP = \angle ADP$.
- (iii) Show that PD bisects $\angle ADE$.
- (iv) If, in addition, $\angle BAP = 90^\circ$ and $\angle APD = 90^\circ$, explain where the centre of the circle is located.

1

2

1

14

Question 4 (12 Marks) Start a new page.

Marks

(a) Find $\int_0^{\frac{\pi}{4}} \frac{e^{\tan x}}{\cos^2 x} dx$

2

(b) By using the “t-results”, where $t = \tan \frac{\theta}{2}$, or otherwise, show that

2

$$\frac{1 - \cos \theta}{\sin \theta} = \tan \frac{\theta}{2}$$

(c) $P(2at, at^2)$ is a point on the parabola $x^2 = 4ay$ with focus $S(0, a)$ and directrix $y = -a$. $D(2at, -a)$ is a point on the directrix.

(i) Display this information on a diagram.

(ii) Show that the equation of the tangent to the parabola at P is
 $y = tx - at^2$

2

(iii) Write down the coordinates of Q, the x-intercept of this tangent.

1

(iv) Show that the tangent is the perpendicular bisector of the interval DS.

2

(d) Use the principle of mathematical induction to show that if x is a positive integer then $(1+x)^n - 1$ is divisible by x for all positive integers $n \geq 1$.

3

Question 5 (12 Marks) Start a new page.

Marks

(a) The area between the curve $y = \sin^2 x$ and the x-axis between $x = 0$ and $x = \frac{\pi}{2}$ is rotated through one complete revolution about the x-axis.

(i) Find the exact value of the area of the region.

3

(ii) Use Simpson's Rule with three function values to find an approximation to the volume of the solid of revolution, leaving the answer in terms of π .

2

(b) A particle moving in a straight line is performing simple harmonic motion. At time t seconds its displacement x metres from a fixed point O on the line is given by $x = 2 \cos^2 t$.

(i) Show that its velocity $v \text{ ms}^{-1}$ and its acceleration $\ddot{x} \text{ ms}^{-2}$ are given by $v^2 = 4(2x - x^2)$ and $\ddot{x} = -4(x - 1)$ respectively.

4

(ii) Find the centre, amplitude and period of the motion.

3

Question 6 (12 Marks) Start a new page.

Marks

(a) Consider the function $f(x) = \frac{x}{4-x^2}$.

(i) Find the domain of the function.

1

(ii) Show that the function is increasing throughout its domain.

2

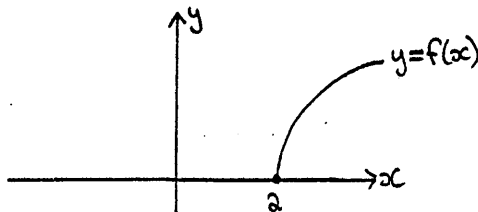
(iii) Sketch the graph of the function showing clearly the coordinates of any points of intersection with the x -axis and the y -axis and the equations of any asymptotes.

3

(b) Take $x = 2.5$ as a first approximation for a root of $x^3 - 3x - 20 = 0$. Use one application of Newton's Method to find a second approximation correct to 2 decimal places.

2

(c) The diagram below shows a sketch of the graph of $y = f(x)$ where $f(x) = \sqrt{x^2 - 4}$ for $x \geq 2$.



(i) Copy this diagram onto your answer sheet. On the same set of axes, sketch the graph of the inverse function $y = f^{-1}(x)$.

1

(ii) State the domain of $f^{-1}(x)$

1

(iii) Find an expression for $y = f^{-1}(x)$ in terms of x .

2

Question 7 (12 Marks) Start a new page.

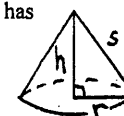
Marks

(a) Evaluate $\lim_{x \rightarrow 0} \frac{\sin\left(\frac{x}{2}\right)}{3x}$

1

(b) A cone with base radius r , vertical height h and slant height s has

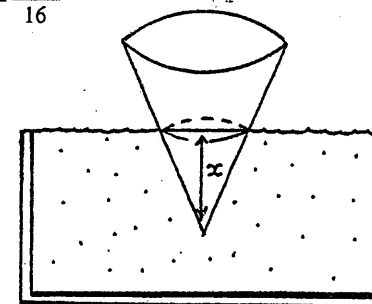
curved surface area $A = \pi rs$ and volume $V = \frac{1}{3} \pi r^2 h$.



(i) Show that a cone with vertical height x and base radius $\frac{3x}{4}$

2

has $A = \frac{15\pi x^2}{16}$ and $V = \frac{3\pi x^3}{16}$



(ii) A similar cone whose base radius is $\frac{3}{4}$ of its vertical height, with its axis

2

vertical and vertex downwards, is being lowered into a container which is overflowing with water. When the vertex is 2 metres below surface of the water (with the cone not fully submerged) the surface area of the cone is

being covered with water at a rate of $0.5 \text{ m}^2 \text{ s}^{-1}$. Show that the cone is being

lowered into the water at a rate of $\frac{2}{15\pi} \text{ ms}^{-1}$.

Question 7 continued over page.

on first page of

16

Question 7 (continued)

Marks

- (iii) Find the rate at which the water is overflowing from the container when the vertex of the cone is 2 metres below the surface.

2

- (c) (i) What is the domain of $h(x) = \sin^{-1} \sqrt{1-x^2} + \sin^{-1} x$?

1

- (ii) Find $h'(x)$

2

- (iii) Hence determine the interval over which $h(x)$ is constant and find this constant.

2

End of paper

1. (a) $\sin^{-1} \frac{x}{5} + c$

(b) $[\ln(1+x) - e^{-x}]_0^1 = \ln 2 - \frac{1}{e} + 1$

© DOMAIN: $-3 \leq x \leq 3$

RANGE: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

$$\textcircled{d} \frac{du}{dx} = 2x \quad \left| \frac{1}{2} \int \sqrt{u} \, du \right.$$
$$\frac{1}{2} du = x dx \quad \left| = \frac{1}{2} \cdot \frac{2}{3} u^{3/2} + C \right.$$
$$= \frac{1}{3} (2+x^2)^{3/2} + C$$

© $x = \frac{-1 \times 5 + 3 \times 1}{2}$, $y = \frac{-1 \times 2 + 3 \times 4}{2}$

$$= -1, \quad = 5$$

i.e. $(-1, 5)$

(f) $m_1 = 2$
 $m_2 = m$ $\tan 45^\circ = \left| \frac{m-2}{1+2m} \right|$

$$\text{i.e. } \frac{m-2}{1+2m} = -1 \text{ or } \frac{m-2}{1+2m} = 1$$

$$m = \frac{1}{3}, -3$$

$$2 \text{ (a) } \frac{-3}{\sqrt{1-9x^2}}$$

$$\textcircled{b} \quad a = \frac{d}{dx} \left(\frac{1}{2} v^2 \right) \\ = \frac{d}{dx} \left[\frac{1}{2} (8x - x^2) \right]$$

$$= 4 - x$$

when $x=3$: $a=1$

③ (i) $u=x$ $v=\ln x$ $y' = 1 - [\ln x + x \frac{1}{x}]$
 $u'=1$ $v'=\frac{1}{x}$

$$= -\ln x$$

$$\begin{aligned} \text{(ii)} \quad \int_1^e \ln x \, dx &= -[x - x \ln x]_1^e \\ &= -(e^2 - e^2 \ln e^2) - (1 - 1 \ln 1) \\ &= e^2 + 1 \end{aligned}$$

$$(d) i) \tau = s + A e^{-kt}$$

$$\frac{dT}{dt} = -k \cdot A e^{-kt} = -k(T-s)$$

(ii) sub. $T=96, S=18, t=0 \Rightarrow A=78$

$$\begin{aligned} \text{sub. } T=75, t=3: 75 &= 18 + 78e^{-3k} \\ 78e^{-3k} &= 57 \\ e^{-3k} &= \frac{57}{78} \end{aligned}$$

$$k = \frac{\ln \frac{57}{78}}{-3} = 0.1046$$

$$\textcircled{iii} \quad 60 = 18 + 78e^{-0.1046t}$$
$$t = 5.9 \text{ min}$$

3. (a) $g(0) = -10$
 $g(2) = 52$] opposite signs
 $\therefore$ root between 0 and 2.

ii) $g(1) = 13 \therefore$ root in 0 to 1.

iii) $g(\frac{1}{2}) = -\frac{1}{8} \therefore$ root in $\frac{1}{2}$ to 1
 $\therefore$ closer to 1.

$$\begin{aligned} \textcircled{b} \frac{d}{dx} (\cos x)^{-1} &= -(\cos x)^{-2} \cdot -\sin x \\ &= \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x} \\ &= \sec x \cdot \tan x \end{aligned}$$

© $dx = du$

$$\begin{aligned}\int \frac{du}{(u-1)^2 + 2(u-1) + 4} &= \int \frac{du}{u^2 + 3} \\ &= \frac{1}{\sqrt{3}} \tan^{-1} \frac{u}{\sqrt{3}} + c \\ &= \frac{1}{\sqrt{3}} \tan^{-1} \frac{x+1}{\sqrt{3}} + c\end{aligned}$$

$$\textcircled{1} \int_{-\frac{\pi}{2}}^0 \sin^2 x \, dx$$

$$i.e. (1+x)^k = Mx + 1$$
 Hence prove true for $n = k+1$
 i.e. $(1+x)^{k+1} - 1$ also divisible
 by x .
 Now $(1+x)^{k+1} - 1 = (1+x)(1+x)^k - 1$

$$= (1+x)(Mx+1) - 1$$
 by our
 assumpt.

$$= x(Mx+M+1)$$
 which is
 divis. by x .
 $\therefore$ if true for $n = k$ then true for $n = k+1$
 Steps: we assumed true for $n = k$ and
 hence proved true for $n = k+1$. Since true
 for $n = 1$ then true for $n = 2$ and so on for
 all the integral values of n .

STEP 2: Assume true for $n=k$.
ie. $(1+x)^k - 1 = Mx$ [M a polynomial]

STEP 1: Prove true for $n=1$.
 $n=1: (1+x)^1 - 1 = x$ which is divisible by x .

∴ TAN. T DS.
Also midpoint DS: $(at, 0)$ which
is Q (on tangent)
∴ Tangent perp. bisector of DS

$$1 - \frac{1}{\gamma} = \frac{1}{\gamma} \Rightarrow \gamma = 2$$

$$\textcircled{!!} y = \frac{x}{a} \quad y' = \frac{1}{a}$$

$$\textcircled{a} \left[e^{\tan x} \right]_0^{\frac{\pi}{4}} = e^1 - e^0 = e - 1$$

continued.

(!!) $\angle s$ in same segment equal.

$\angle PDE = \angle PBC$ (ext. $\angle$ cyclic quad.

$DCBP$ equals opp. interior $\angle$)

$\angle APD = \angle PBC$ (both equal $\angle ABP$)

$\therefore \angle PDE = \angle APD$

i.e. PD bisects $\angle ADE$

Intersection of AD and BP

L in semi-circle 90° means both are diameters.

continued

$$i) \div \pi \times \frac{\pi}{3} \left[0 + 1 + 4 \times \left(\frac{1}{\sqrt{2}} \right)^4 \right]$$

$$= \frac{\pi^2}{6} \text{ units}^3$$

$$ii) x = 2(\cos t)^2$$

$$\dot{x} = 4 \cos t \cdot -\sin t = -4 \cos t \cdot \sin t$$

$$v^2 = 16 \cos^2 t \sin^2 t$$

$$= 16 \cos^2 t (1 - \cos^2 t)$$

$$= 16 \times \frac{x}{2} \left(1 - \frac{x}{2} \right)$$

$$= 4(2x - x^2)$$

$$\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$= \frac{d}{dx} (4x - 2x^2)$$

$$= 4 - 4x$$

$$= -4(x - 1)$$

CENTRE: $\ddot{x} = 0 \therefore$ CENTRE $x = 1$

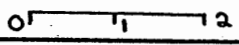
EXTREMITIES: $v = 0 \therefore 2x - x^2 = 0$

$$x(2 - x) = 0$$

$$x = 0, 2$$

$\therefore$ AMPLITUDE = 1

PERIOD = $\frac{2\pi}{n} = \frac{2\pi}{2} = \pi$



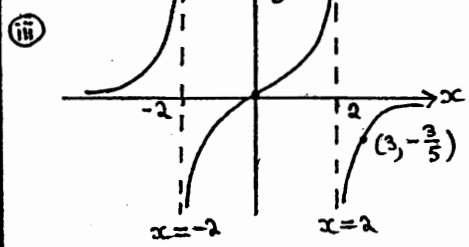
6. a) i) all real $x, x \neq \pm 2$

ii) Find $\frac{dy}{dx}$: $u = x \quad v = 4 - x^2$
 $u' = 1 \quad v' = -2x$

$$\frac{dy}{dx} = \frac{(4 - x^2) \cdot 1 - x \cdot (-2x)}{(4 - x^2)^2}$$

$$= \frac{4 + x^2}{(4 - x^2)^2} > 0 \text{ for all } x \text{ in domain}$$

$\therefore$ increasing



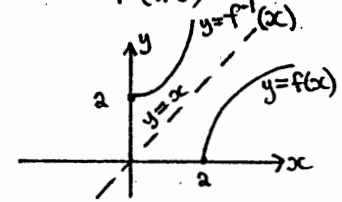
b) $P(2.5) = -11.875$

$$P'(x) = 3x^2 - 3$$

$$P'(2.5) = 15.75$$

$$Z_2 = 2.5 - \frac{P(2.5)}{P'(2.5)} = 3.25$$

c) i)



ii) domain: $x \geq 0$

iii) $x = \sqrt{y^2 - 4}$

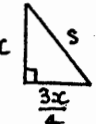
$$x^2 = y^2 - 4$$

$$y^2 = x^2 + 4$$

$$y = \pm \sqrt{x^2 + 4} \quad [\text{TAKE +VE ONLY}]$$

$$y = \sqrt{x^2 + 4}$$

7. a) $\lim_{x \rightarrow 0} \frac{\sin(\frac{x}{a})}{\frac{x}{a}} \times \frac{1}{6} = 1 \times \frac{1}{6} = \frac{1}{6}$

b) i) 

$$s = \sqrt{x^2 + \left(\frac{3x}{4} \right)^2}$$

$$= \frac{5x}{4}$$

$$A = \pi \times \frac{3x}{4} \times \frac{5x}{4} = \frac{15\pi x^2}{16}$$

$$V = \frac{1}{3} \times \pi \times \left(\frac{3x}{4} \right)^2 \times x = \frac{3\pi x^3}{16}$$

ii) We want $\frac{dx}{dt}$. We know $\frac{dA}{dt} = 0.5$

$$\frac{dA}{dt} = \frac{dA}{dx} \times \frac{dx}{dt}$$

$$0.5 = \frac{15\pi x}{8} \times \frac{dx}{dt}$$

$$0.5 = \frac{15\pi}{4} \times \frac{dx}{dt} \text{ when } x = 2$$

$\therefore h(x)$ is not constant for $x < 0$.

$$\frac{x-1}{2} =$$

$$\frac{x-1}{1} + \frac{x-1}{-x} = h'(x)$$

N.B. For $x < 0$:

$\therefore$ constant for $0 < x \leq 1$ and this constant is $\frac{\pi}{2}$

and $h(0) = h(1) = \frac{\pi}{2}$
 But domain is $-1 < x \leq 1$
 $\therefore h(x)$ is constant for $x > 0$

so for $x > 0$:

$$\frac{x-1}{1} + \frac{x-1}{-1} =$$

$$\frac{x-1}{1} + \frac{x-1}{-x} = h'(x)$$

Now $\sqrt{x^2} = x$ for $x \geq 0$
 $= -x$ for $x < 0$

$$\frac{x-1}{1} + \frac{x-1}{-x} =$$

$$\frac{x-1}{1} +$$

$$\frac{x-1}{-x} \times \frac{x(x-1)-1}{1} = h'(x)$$

iii) $1 \leq x \leq 1$
 i.e. overflowing at rate of $0.3 \text{ m}^3/\text{s}$

$$0.3 =$$

$$\frac{16}{36\pi} \times \frac{16}{15\pi} =$$

$$\frac{16}{9\pi x^2} \times \frac{16}{15\pi} =$$

$$\frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dt}$$

iii) We want $\frac{dV}{dt}$

ii) $\frac{dx}{dt} = \frac{15\pi}{2}$ i.e. lowered at rate of $\frac{15\pi}{2} \text{ ms}^{-1}$

7. continued